# Broadband Expansion and Small-Firm Productivity: Evidence from a Staggered Municipal Rollout

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*Demonstration manuscript; fictional setting and simulated data.*

# Key question

**Can municipal broadband availability raise small-firm productivity?**

**Why this matters:**

- Small firms adopt new technologies later and less completely
- Distance can limit access to specialized providers and inputs

**Empirical test:**

- Is labor productivity higher after municipal service arrives?
- Is the difference larger for communication-intensive firms?

**Policy relevance:** whether universal infrastructure changes small-firm production.

# This paper

We link 9,400 firms in 2008–2019 to broadband rollout across 284 municipalities.

1. **Each 2011–2015 rollout cohort is more productive after backbone arrival.**
   - Pooled two-way estimate: 0.034 log points, a weighted summary (backup)
2. **Communication-intensive firms show a larger difference.**
   - One SD adds 0.0064 log points, about one-fifth of the pooled difference
   - The model links this pattern to the value of remote coordination

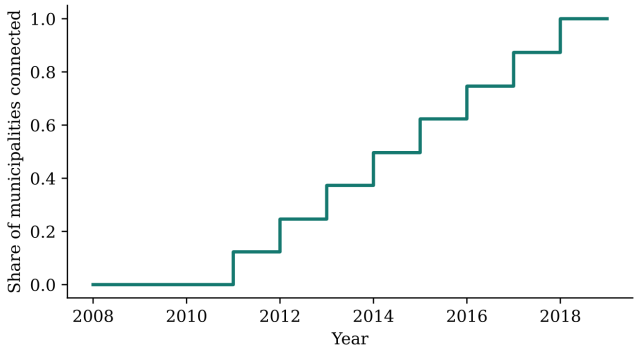
**Implication:** the differences track communication intensity, as the model predicts.

# Roadmap

1. Setting and data
2. Design and main evidence
3. Model and interpretation

# Broadband reached every municipality in eight waves

*All 284 municipalities connected between 2011–2018*




- **Measurement:** exposure begins with municipal service; firm adoption is unobserved.

# Data

*Business Register, audited accounts, and employer–employee records, 2008–2019*

- **Sample:** 9,400 firms and 112,800 firm-years, 2008–2019
  - 3–49 employees in 2008, positive sales, balanced survivor panel
- **Outcome:** real value added per worker
  - audited accounts deflated by industry prices, divided by FTE employment
- **Broadband:** first commercial-service year in the headquarters municipality
- **Communication intensity:** 2008 wage-bill share in communication-intensive jobs

▸ Sample details

# The design follows firms as local backbone service arrives

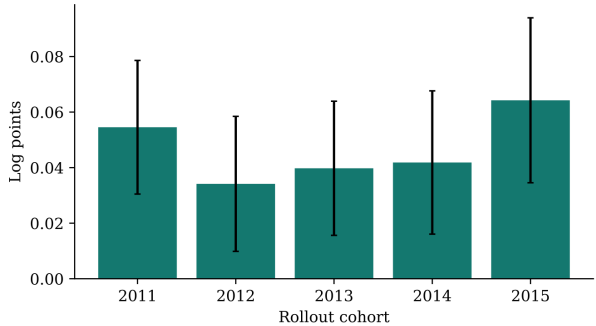
*Within-firm comparison across different municipal arrival dates*

$$y_{it} = \alpha_i + \lambda_t + \beta B_{m(i)t} + X'_{it}\delta + \varepsilon_{it}$$

- **Equation:** pools all timing comparisons, including already-connected controls
- **Main evidence:** each 2011–2015 wave versus places connected 3+ years later
- **Counterfactual:** municipalities served at different dates would follow similar paths
- **Diagnostic:** pre-arrival coefficients check for diverging trends before arrival

# Each 2011–2015 cohort is more productive after arrival

*Paper Figure 3; versus places connected 3+ years later (event window not stated)*

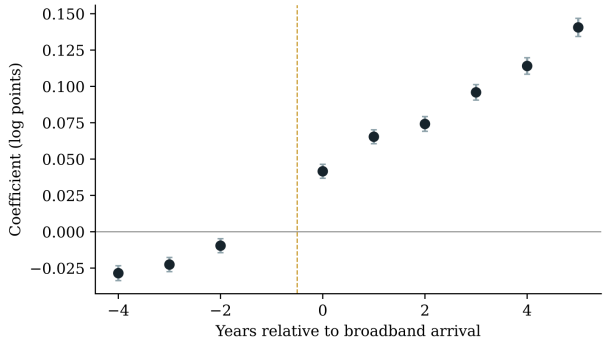



- **Reading:** every wave shown is higher after arrival; no single wave drives the sign.

▸ Pooled estimate ▸ Source notes ▸ Inference

# Productivity rises before and after broadband arrives

*Paper Figure 2; two-way event study; $t = -1$ omitted; firm-clustered 95% intervals*




- **Limitation:** The visible pretrend limits the estimates to a descriptive reading.

▸ Assignment ▸ Placebo

# The model predicts larger gains for communication-intensive firms

*Broadband can make specialized remote providers worthwhile*

**Switching case:**

$$\kappa_1 p \leq \phi w < \kappa_0 p, \qquad G(\theta_i) = \eta \ln[1 + (\phi - 1)\theta_i]$$

$$G'(\theta_i) = \frac{\eta(\phi - 1)}{1 + (\phi - 1)\theta_i} > 0$$

- $\theta_i$ is firm $i$'s pre-program communication intensity
- $\kappa$ is the cost of coordinating with remote providers
- $\phi > 1$ is the quality advantage of specialized remote providers

**Prediction:** communication-intensive firms can shift more tasks to remote providers.

▸ Derivation

# The broadband association is larger for communication-intensive firms

*Communication intensity is fixed at its 2008 value*

| | With interaction | Average only |
|---|---|---|
| Broadband | 0.020*** | 0.034*** |
| | (0.003) | (0.002) |
| Broadband × communication intensity | **0.032*** ** | |
| | (0.005) | |
| Firm and year effects | Yes | Yes |
| Observations | 112,800 | 112,800 |
| One SD: $0.032 \times 0.20 =$ **0.0064** log points | | |

Firm-clustered SEs; *** $p < 0.01$ as in the source; communication-intensity SD = 0.20 (Table 1).

- The larger gap for communication-intensive firms matches the model.

▸ Interpretation

# Conclusion

- **Each 2011–2015 cohort is more productive after broadband arrives.**
- The difference is larger for communication-intensive firms.

**Implication:** the differences track communication intensity, as the model predicts.

# Appendix

# The balanced panel describes surviving small firms

*Paper Section 4.2 and Table 1, p. 5*

- **Population:** registered firms linked to audited accounts and worker records
- **Initial screen:** 3–49 employees in 2008; positive sales; at least four consecutive years
- **Final restriction:** a balanced 2008–2019 panel
- **Delivered sample:** 9,400 firms and 112,800 firm-years
- **Scope:** the comparison applies to selected survivors; entry and exit are excluded

**Source issues:** balance rule unexplained; reported employment mean 23.4 exceeds maximum 19.

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# Projected growth helped determine rollout order

*Published RDDF priority rule, paper Section 4.1*

- **Coverage commitment:** the 2010 Act promised universal connection by 2018
- **Priority rule:** earlier service for low-productivity, high-growth municipalities
  - below-median productivity levels
  - above-median projected growth in internal assessments
- **Engineering constraints:** distance and terrain also affected construction
- **Unresolved:** the paper does not isolate this variation from the growth score

**Identification requires:** arrival timing unrelated to counterfactual productivity trends.

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# The pooled two-way estimate sits at the low end

*Paper Table 2, p. 7; log labor productivity, firm and year fixed effects*

| | Baseline | Firm controls | Region trends | Municipal SEs |
|---|---|---|---|---|
| Broadband | **0.034*** ** | 0.033*** | 0.030*** | 0.034*** |
| | (0.002) | (0.002) | (0.002) | (0.003) |
| Firm and year effects | Yes | Yes | Yes | Yes |
| Observations | 112,800 | 112,800 | 112,800 | 112,800 |

SEs are firm-clustered except the final column (municipality-clustered); *** $p < 0.01$, as in the source.

- **Estimator:** a weighted mix of all timing comparisons, it can understate rising effects.

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# Rounded placebo entries cannot confirm the prose

*Pre-treatment sample; rollout years permuted across municipalities*

| | Pre-period placebo |
|---|---|
| Permuted broadband | −0.004 |
| | (0.002) |

- The coefficient-to-SE ratio is about −2; rounding allows −1.4 to −3
- The prose calls the estimate statistically indistinguishable from zero
- The source provides no exact *p*-value or confidence interval

- **Unresolved:** the rounded entries fit both a zero and a nonzero reading.

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# Two numerical inconsistencies require correction

*Narrative prose versus exact table cells*

| Quantity | Narrative source | Exact table source |
|---|---|---|
| Headline magnitude | Abstract: 34%; text: 3.4% | Table 2: 0.034 log points |
| Region-trend estimate | Section 6.1: 0.024 | Table 2: 0.030 |

- The talk reports the coefficient's table unit: log points
- The pooled-estimate backup uses exact Table 2 cells in every specification
- A 0.034 log-point difference is approximately 3.4%, not 34%

- **Correction:** the exact cell is 0.034 log points; both narrative values need revision.

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# Table 5 is stable but not self-consistent

*Exact reported cells, paper p. 9*

| | Region trends | No controls | Drop 2011–12 |
|---|---|---|---|
| Broadband | 0.030*** | 0.034*** | 0.037*** |
| | (0.002) | (0.002) | (0.002) |

- **Stable pattern:** reported coefficients range from 0.030 to 0.037
- **Header–note conflict:** "No controls" versus "full control set"
- **Prose–table conflict:** 0.024 versus 0.030 for region trends

- **Reading:** the coefficient range is narrow, but the source labels need correction.

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# Municipal assignment calls for municipal-level inference

*Treatment timing varies across 284 municipalities*

| | Firm-clustered | Municipality-clustered |
|---|---|---|
| Broadband | 0.034 | 0.034 |
| | (0.002) | (0.003) |

- The baseline clusters by firm, below the municipal assignment level
- Table 2 column (4) reports the assignment-aligned alternative
- The point estimate is unchanged and the reported SE rises to 0.003

- **Inference:** clustering changes uncertainty, not the concern about selected timing.

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# Larger gains require tasks to switch providers

*Paper Proposition 1 and proof, pp. 2–3*

$$\underbrace{\kappa_1 p}_{\text{remote cost after}} \leq \underbrace{\phi w}_{\text{quality-adjusted in-house cost}} < \underbrace{\kappa_0 p}_{\text{remote cost before}}$$

$$G(\theta_i) = \eta \ln[1 + (\phi - 1)\theta_i], \qquad G'(\theta_i) = \frac{\eta(\phi - 1)}{1 + (\phi - 1)\theta_i} > 0$$

- Before arrival, communication-intensive tasks remain in-house
- After arrival, the same tasks switch to specialized remote providers
- The derivative is positive because $\eta > 0$ and $\phi > 1$
- **Scope:** the heterogeneity prediction is conditional on this switch

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# The interaction is consistent with, not unique to, outsourcing

*What Table 4 observes and what it leaves unmeasured*

- **Observed:** a positive interaction with 2008 communication intensity
  - one SD adds 0.0064 log points
- **Predicted:** the task model gives the same positive sign
  - the switching case makes $G'(\theta_i) > 0$
- **Unobserved:** remote procurement, task reorganization, and channel-specific inputs
- **Alternative:** other firm traits can generate the same pattern

**Discriminating evidence:** direct measures of sourcing before and after arrival.

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# The six-year payback is not traceable from the reported inputs

*Paper Section 7, pp. 8–9*

- **Claimed flow:** 410 million euros per year against 2.3 billion euros in construction
- **Estimate:** a productivity association among sample survivors
- **Missing inputs:** firm value added, program coverage, and maintenance costs
- **Missing assumptions:** causality, persistence, discounting, and incidence
- **Unit problem:** gross output is not a fiscal return

**Conclusion for this talk:** the payback and cost-effectiveness claims are excluded.

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